

Fault detection in insulators based on ultrasonic signal processing using a hybrid deep learning technique

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Abstract: Identifying problems in insulators is a task that requires the experience of the operator. Contaminated insulators generally do not represent a system failure, however, due to higher surface conductivity, they may suffer from electrical discharges and may result in irreversible failures. The identification of possible failures in inspections can help to forecast faults to improve reliability in the power grid. Based on this need, this article presents a study on fault prediction in distribution insulators, through a laboratory evaluation in a contaminated insulator, where 13.8 kV (root mean square) was applied considering an ultrasound detector connected to a computer for data set acquisition. In the sequence, a time series prediction, using a hybrid deep learning technique defined as wavelet long short-term memory (LSTM), was performed. The hybrid LSTM was proposed considering feature extraction through the wavelet energy coefficient. Finally, for a complete evaluation, deeper LSTM layers were included, and both the training method and the hardware configuration were evaluated. The wavelet LSTM algorithm showed interesting accuracy results when compared to classic prediction algorithms like the non-linear autoregressive exogenous model.

1 Introduction

The main function of the electric power system is to deliver electrical energy to customers as economically as possible, always considering an acceptable degree of quality associated with the service [1]. During the past years, the use of renewable energy sources (RESs) has increased, mainly due to the growing prices of conventional energy supplies. Considering the actual panorama of energy generation, a reliable electrical distribution system became the primary requirement of the smart grid [2], especially because power fluctuations caused by RESs can affect the stability, frequency control, and reliability of the power system [3]. To maintain the electrical network in operation, it is necessary to identify faults and correct them before irreversible faults occur.

Insulators are important components of the power grid, as they have the function of supporting cables and keeping the system isolated. When an insulator is defective, it loses its insulation capacity and can cause power system failures. To keep the electric power system in operation, it is essential to meet the reliability requirements of the inspection agency [4]. Moreover, to provide a reliable system preventive maintenance is necessary, and a contaminated insulator may indicate that there will be a failure in the future. The contamination increases the insulator surface conductivity and thus facilitates the possibility of local discharges [5].

Discharges can generate tread or erosion on the insulator surface, when this happens the insulator loses its insulation characteristics, becoming more susceptible to disruptive discharges. A good fault prediction system can detect degradation along the time of insulating components and thus indicates future vulnerable points in the distribution network [6].

The inspection of the electrical system based on either pattern recognition or prediction of failures became an important issue for utility companies, and critical for the grid to remain operational [7–9]. In this way, the application of advanced techniques based on artificial intelligence (AI) can automate inspections and increase reliability in the electric power system [10].

By analysing traditional fault diagnosis methods, it can be found a lack of adaptive processing associated with dynamic information. In this way, it can be noticed that simplified strategies suffer to solve feature extraction tasks [11]. Experimental results show that the long short-term memory (LSTM) can classify different faults and provide more promising performance in fault diagnosis [12].

Early fault detection is crucial for reducing the downtime of a process or system. According to [13], the LSTM has better accuracy results for fault classification than classic multilayer perceptron (MLP) and support vector machine (SVM) for the chemical process. In this research, the proposed method for fault diagnosis is without specific feature extraction. The accuracy reached 98.7%, showing that the LSTM method is an interesting approach to solve the problem.

The LSTM can discover temporal dependencies in sequential data, which allows the estimator to make use of previous context information and makes it more robust when distortions appear along the time axis. Using LSTM for predicting industrial machine failures, in the work presented in [14], it was obtained good assertiveness when detecting 566 faults through a deep neural network. Their study presented a complete descriptive evaluation with LSTM, SVM, among other algorithms.

Based on the need for reliability improvement associated with the electrical system, Bhattacharya and Sinha [15] created an AI

system to analyse the power grid information and determine their health, reaching 97% accuracy, showing up to a 6% improvement concerning SVM classifier. For this, several conditions of the electric network were simulated. Data were used for both training and validation assuming an LSTM model. According to the authors, the method provided excellent accuracy for their problem and the technique can be used for more complex applications.

For fault prediction in power systems, Zhang *et al.* [16] developed similar research, where the LSTM model performed well when extracting time series features over a long period. The work was based on data from the Wanjiang substation in China Southern Power Grid. A practical presentation showed the feasibility of applying LSTM in real scenarios, mainly because of the algorithm's ability to handle large time series. The paper presented a complete description of the LSTM network, showing that the algorithm is superior to SVM, backpropagation neural networks, and stacked autoencoders.

LSTM applications in prediction or fault identification tasks, specifically on insulators, are rare. One example can be found in [17] using LSTM for insulator fault identification, where an image recognition and processing model for electrical inspection was proposed. The model considered distinct information sources such as time, position, geography, and climate to help utilities with decision making. The research also assumed insulators and transmission line images gathered by patrol robots and unmanned aerial vehicles. Finally, a natural language processing system was developed to describe the images, helping maintenance operations. Using LSTM, presented promising results compared with other classic algorithms, such as SVM, MLP, dynamic linear discriminant analysis, and hybrid algorithms using dynamic principal component analysis.

The research developed in [18] presented an analysis of the life expectancy of composite insulators in high temperature of humidity regions based on physical and statistical models, considering components that are in the high voltage network for a long time. The service life of the sampled insulators was determined by the thermal oxidation process of the silicone rubber, mechanical load tests, and spectroscopy. The study was based on a large volume of samples taken from the power transmission system. Despite a fairly complete evaluation in terms of material characteristics, the statistical technique used for forecasting insulators conditions is simple, which did not capture non-linearities available on physical data.

Discharges in transmission insulators are a growing problem in the power system as there is a tendency to use higher voltage levels [5]. A probability forecasting of high-risk events was presented in [19] to predict leakage current values in high voltage insulators. In this evaluation, the calculation of the probability density of leakage current by kernel density estimator presented relevant results that ensure the safe operation of the power system. The application of probability density function may be a viable alternative for the evaluation of leakage current forecasting in high voltage insulators.

Electrical discharges on insulators can occur in different ways, from partial discharges to continuous electrical arcs as presented in [20], which specifically discusses the problem. The previously cited study evaluated different forms of surface discharges in contaminated insulators. During inspections and fault identification, the ultrasound equipment stands out due to its directional characteristic, which can be used to identify the correct place where a specific fault occurs in the electrical system.

The prediction of faults in insulators is an important issue, considering that the proper functioning of these components helps to increase the reliability in the electrical energy supply. Based on this assumption, as emphasised before in considerable studies, LSTM is one of the most advanced techniques that can be used for fault prediction based on chaotic time series. Additionally, the application of the wavelet technique for feature extraction can further improve this analysis by reducing noise and increasing the algorithm accuracy [21].

In this sense, this article presents a failure prediction assessment based on the ultrasonic signal emitted by a contaminated insulator that was removed from the electrical system during network inspection. To solve the problem, using a quantitative methodology

[22], we considered a hybrid deep learning technique defined as wavelet LSTM. The algorithm, which combined two techniques, uses wavelet for feature extraction, and the LSTM to evaluate the prediction of fault development. For a complete algorithm analysis, variations of the training method, the amount of data used for training, and deeper layer capacities were also considered during the algorithm evaluation.

2 Overview of insulators condition

Electrical distribution and transmission systems have insulators that are generally outdoors [23]. In the outdoor environment, insulators are exposed to climate variations and diverse environmental conditions. Coastal power grids generally have a higher salinity accumulation on their surface, power grids that are located near unpaved roads may also have a higher accumulation of contaminants [24].

Surface contamination of insulators can increase surface conductivity and make components more vulnerable to electrical discharges. To prevent the electrical system from shutdowns, those caused by insulation failures, the electric utility develops different types of inspections for the electrical distribution branches [25].

During inspections, it is possible to determine those components with higher chances of developing problems and/or to damage the electrical system. In a field inspection, a team usually goes through the problematic extension using specific equipment, such as ultrasonic, camera, ultraviolet, and infrared tools, aiming to find signs of failure [26].

Among the equipment used during inspections, stands out the ultrasonic monitor apparatus. Due to its directional characteristic, the ultrasound equipment can indicate the problematic insulator among many others. Focusing on the specific equipment adopted in this research, the defective insulator generates ultrasound signals that are converted into an audible signal by the equipment. This sound has to be interpreted by the operator to identify possible faulty components [27].

The inspection of the electrical system has been improved over the years through techniques and equipment capable to automate the operator's tasks. The defects location is difficult to be accomplished, mainly justified by the large extension of the electric power branch to be analysed, and because failures are usually not visible by inspections from the ground. One aspect that is important to address is that, after a fault, the line protection normally recloses successfully, which makes finding the fault difficult [28]. However, the defect is still there, and the system becomes more vulnerable and may gradually lose its reliability.

The use of AI for fault identification and fault prediction has been attracting attention in the search for solutions to improve grid reliability [16]. Considering that the signal captured by the ultrasound detector can be stated as a time series, AI methods can be assumed to predict failures based on time series forecasting approaches.

Contaminated insulators do not represent a system failure that ensures the component replacement action, however, it may be the indication that a failure will develop on-site [29]. In this way, through AI applied for forecasting purposes, it is possible to predict the development of failures based on the ultrasound signal from a contaminated insulator.

The question of whether the signal can change is related to the non-linear characteristic of the fault. For this reason, fault identification is a difficult task, and models based on AI, like the one presented in this article, can improve the ability to identify distinct types of faults or predict them.

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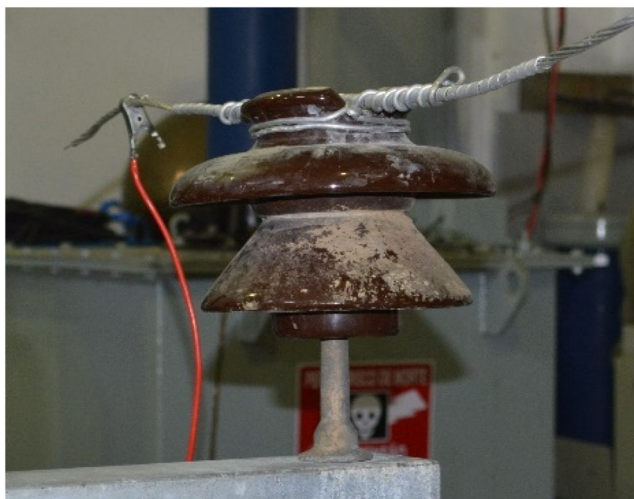


Fig. 1 Contaminated ceramic insulator removed during field inspection and mounted in the laboratory [30]

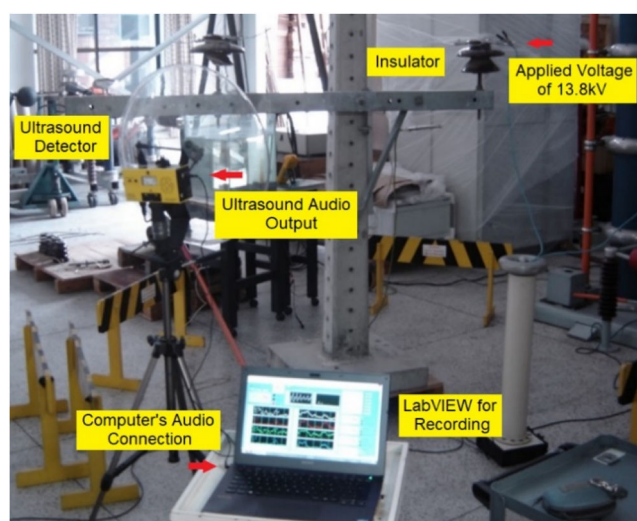


Fig. 2 Laboratory setup

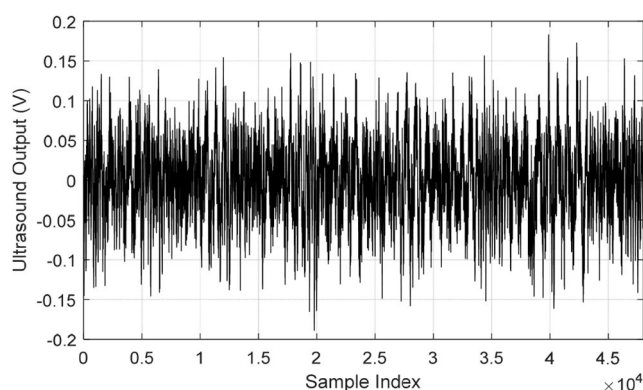


Fig. 3 Recorded signal of the contaminated ceramic insulator

3 Data acquisition: method and experiment

This section presents the characteristics of the sample used in this study and the laboratory setup assumed for data acquisition. The mooring and fixation of the components were performed according to the standards of the local electricity utility *Centrais Elétricas de Santa Catarina* (CELESC – Brazil).

3.1 Contaminated insulator sample

Based on the condition reported in the previous section, the insulator considered as a sample is presented in Fig. 1. This insulator was removed from the distribution system after an

electrical inspection that was requested by the electric utility, as there were records of shutdowns at the power supply branch. The place is in a rural region, which has unpaved roads near the power grid [31].

As it can be observed in the sample, there were residues accumulation on the surface of the component, such as dust and organic contamination. At some specific insulators, the contamination is strongly stuck and difficult to remove [30].

3.2 Laboratory setup

The sample was mounted in the laboratory using the same pattern assumed in the field. The experiment was set up on a 2 m long crosshead on a 1.5 m high pole. The ultrasound equipment was placed on a fixed tripod 1 m high; this height was assumed because it is equivalent to that used during the field inspection by the operator. This equipment was positioned at a distance of 2.2 m away from the sample, the distance can reduce the intensity of the measured signal, for this reason, it is necessary to adjust its sensitivity. The equipment is directional there is a need to be correctly pointed to measure the signal.

A nominal voltage of 13.8 kV [root mean square (RMS)], 60 Hz, was applied, which is the same used in feeders in rural areas in southern Brazil. The ultrasound detector from Radar Engineers® model 250 was used for measurements, which is the same equipment adopted in field inspection.

The ultrasound equipment used in this work was developed specifically for fault location in the electric power system and has been used for several years by electric power companies to identify damaged components. With the accumulation of contaminants on the insulators' surfaces, failures may occur. In this sense, a contaminated insulator can be identified through the characteristics of the ultrasound signal, considering that partial or disruptive discharges can be previously detected by this equipment.

The audio output of the ultrasound detector was connected to the computer's audio input for online data acquisition, as seen in Fig. 2. A computational code using LabVIEW software was developed for this purpose. To avoid interference noise, we turn off all electrical equipment, including lighting to record the signal.

The data used for both training and validation phases of the deep LSTM model are those recorded from the ultrasound equipment. The signal produced by this equipment is based on a chaotic time series. This signal has this characteristic considering that the discharges generated during the experiment are non-linear events that must be identified by the operator to find faults in the components being evaluated.

The ultrasound signal was recorded at a frequency rate of 48 kHz, i.e. with a sampling rate of 2.08×10^{-5} s. This frequency was chosen because the equipment was designed to generate a signal in the audible range (20 Hz–20 kHz), and thus using the 48 kHz frequency the signal was fully recorded. The ultrasound equipment used was designed to detect the noise spectrum of spark sources from 1 kHz–1 GHz. The recorded signal can be seen in Fig. 3.

4 Deep time series forecasting strategy

In this section, the technique assumed to predict chaotic time series is addressed. To consider LSTM networks for time series forecasting purposes, the values of future samples of a sequence are shifted by a time step. With this, at each input sequence, the LSTM learns how to predict the value associated with the time series for the next time step.

A brief introduction about LSTM is presented in the sequence, followed by the algorithm configuration, and the technique assumed for feature extraction.

4.1 Long short-term memory

The LSTM is an artificial recurrent neural network (RNN) frequently used in deep learning applications. The LSTM has feedback connections, unlike the feed-forward neural networks. An LSTM unit usually is composed of a cell, an input gate, an output gate, and a forget gate [32]. The gates are responsible for removing or adding information to the cell state.

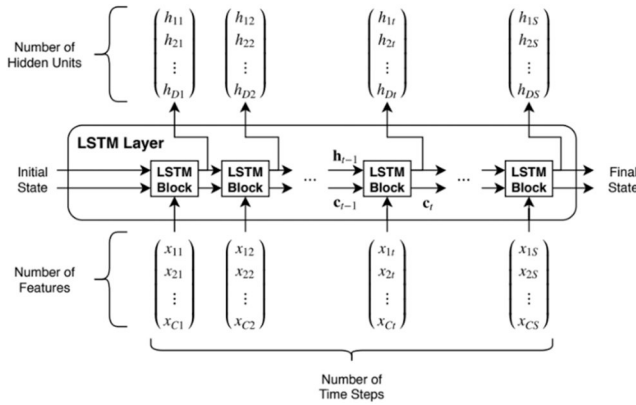


Fig. 4 Flow of a time series for LSTM [34]

The main concept behind an LSTM is that it was developed to deal with gradient problems found during the training of the RNNs. In this way, based on time series data, an LSTM is well-suited to make predictions. Moreover, relative insensitivity to gap length is an advantage of LSTM over alternative RNNs, hidden Markov models, and other sequence learning methods [3].

LSTMs can remember the state of the network between forecasts, this is useful when there is no complete time series in advance or it is desired to make a prediction n -steps ahead in a time series [33]. The flow of a time series x with resources c of length s through an LSTM layer is shown in Fig. 4, where S is the number of time steps, C is the number of features and D is the number of hidden units.

In Fig. 4, h_t and c_t denote the output and state of the cell at a time step t , respectively. The first LSTM block uses the initial network state and the first sequence to calculate the first output and to update the cell state. In a time interval t , the block uses the current state of the network (c_{t-1} , h_{t-1}) and the next sequence to calculate the output and updated cell state c_t .

The key in LSTM structure is the cell state, also known as a memory cell [34]. As can be verified from Fig. 4, an input at time t and the hidden state from the previous time steps ($t-1$) are introduced in the LSTM block, and then the hidden state is computed. The first step inside the LSTM block is to decide what information is going to be thrown away from the cell state. The next step decides which new information is going to be stored in the cell state. Then, the LSTM block updates the old cell state into the new cell state.

To conclude the operation inside the block, the structure decides what is going to be the produced output based on the cell state. By associating many LSTM blocks (Fig. 4), those connected in a deep recurrent network, it is possible to combine the advantages of a single LSTM layer to build features at the lower layers that will disentangle the factors of variations in the input data, combining these representations at higher layers [35].

The state of the layer is based on both the hidden state and the state of the cell. The hidden state in time step t contains the output of the LSTM layer for this time interval. The cell state contains information learned in the previous time steps. At each time step, the layer adds information or removes information from the cell state. The layer controls these updates using gates.

The learned weights of an LSTM model are the input weights: W , the recurrent weights: R , and the bias: b . The vectors W , R , and b are the concatenations of the input weights, recurrent weights, and trends of each component, respectively, as defined in

$$W = \begin{bmatrix} W_i \\ W_f \\ W_g \\ W_o \end{bmatrix}, R = \begin{bmatrix} R_i \\ R_f \\ R_g \\ R_o \end{bmatrix}, b = \begin{bmatrix} b_i \\ b_f \\ b_g \\ b_o \end{bmatrix} \quad (1)$$

where i denotes the input gate to control the level of cell state update, f is the forget gate for state reset control, g is the cell candidate to add information, and o is the output gate that controls

the level of cell state added to hidden state. The following equations describe the components at time step t .

$$\begin{aligned} i_t &= \sigma_g(W_i x_t + R_i h_{t-1} + b_i), \\ f_t &= \sigma_g(W_f x_t + R_f h_{t-1} + b_f), \\ g_t &= \sigma_c(W_g x_t + R_g h_{t-1} + b_g), \\ o_t &= \sigma_g(W_o x_t + R_o h_{t-1} + b_o). \end{aligned} \quad (2)$$

In the previous equations, σ_g denotes the gate activation function. In this work, σ_g was assumed as a sigmoid function:

$$\sigma_g(t) = \frac{1}{1 + e^{-t}}. \quad (3)$$

The cell activation function, σ_c , is used to update both cell and hidden states representing the hyperbolic tangent function:

$$\sigma_c(t) = \frac{e^t - e^{-t}}{e^t + e^{-t}}. \quad (4)$$

To evaluate the model, the LSTM regression network was specified according to the variation of both the hidden units and the units in the deeper layers. In this way, we define the algorithm as a deeper LSTM. Additionally, three forms of training were evaluated, stochastic gradient descent with momentum (SGDM), RMS propagation (RMSProp) optimiser, and adaptive moment estimation (ADAM) [36].

The stochastic gradient descent algorithm updates the network parameters to minimise the loss function by taking small steps in each iteration toward the negative loss gradient,

$$\theta_{i+1} = \theta_i - \alpha \nabla E(\theta_i). \quad (5)$$

In (5), i is the number of iterations, α is the learning rate, θ is the parameter vector, and $E(\theta)$ represents the loss function. In each iteration, the stochastic algorithm evaluates the gradient and updates the parameters using a subset of training data.

The stochastic method can oscillate along the steepest descent path towards the optimum value. Adding a momentum term to the parameter update procedure, it is possible to reduce the oscillations. Finally, the parameter vector is updated considering a momentum (SGDM) given by

$$\theta_{i+1} = \theta_i - \alpha \nabla E(\theta_i) + \gamma(\theta_i - \theta_{i-1}), \quad (6)$$

where γ determines the contribution of the previous gradient step to the current iteration. The SGDM method uses a single learning rate for all the parameters.

The RMSProp training uses learning rates that differ by parameter and can automatically adapt to the loss function being optimised. Thus, the algorithm keeps a moving average v of the squares of the elements based on the parameter gradients and can be calculated as

$$v_i = \beta_2 v_{i-1} + (1 - \beta_2) [\nabla E(\theta_i)]^2. \quad (7)$$

In this way, β_2 is the decay rate of the moving average. The corresponding average lengths of square gradients are equal to $1 / (1 - \beta_2)$. This algorithm uses this moving average to normalise the updates of each parameter individually,

$$\theta_{i+1} = \theta_i - \frac{\alpha \nabla E(\theta_i)}{\sqrt{v_i + \epsilon}}. \quad (8)$$

By assuming RMSProp, the algorithm effectively reduces learning rates for large gradient parameters and increases learning rates for small gradient parameters. The ϵ constant is used to avoid division by zero.

Finally, the ADAM approach maintains an elementary moving average m of both gradient parameters and their square values, with the gradient decay rate denoted by β_1

$$m_i = \beta_1 m_{i-1} + (1 - \beta_1) \nabla E(\theta_i). \quad (9)$$

If the gradients on consecutive iterations are similar, the use of a moving average gradient allows updating the parameters capturing the moment of a certain direction.

$$v_i = \beta_2 m_{i-1} + (1 - \beta_2) [\nabla E(\theta_i)]^2. \quad (10)$$

The ADAM method uses moving averages to update the network parameters as:

$$\theta_{i+1} = \theta_i - \frac{\alpha m_i}{\sqrt{v_i} + \epsilon}. \quad (11)$$

The adaptive learning for all models was assumed considering the initial learning rate of 0.005, and progressively multiplied by 0.2.

In this work, the authors evaluated the network performance using the central processing unit (CPU) of a personal computer. For comparison purposes, the graphics processing unit (GPU) and the automatic (AUTO) mode, where processing was performed based on hardware availability, were also considered. The entire analysis was performed considering an Intel Core i7-3520M, equipped with 8 GB of random-access memory, and a GeForce GT 640M NVIDIA GPU video card with CUDA enabled. Finally, the MATLAB® software was used during code development.

4.2 Algorithm configuration

To predict the values of the future time steps of a time series using an LSTM, it is necessary to train the model using sequence regression. To perform this task, data was partitioned in training and validation subsets. In this work, it was considered 50, 75, and 90% of the data set for training. Thus, cross-validation was performed, the training data set was not used during the validation phase of the network. The data used for training the network were not used for validation of the algorithm, thus being cross-validation by hold-out approach. The total signal length is 48,000 samples, which were divided for the evaluation of the algorithm.

For better fit and to avoid divergent training, the training data were standardised to zero mean. The analysis was based only on training and validation procedures. The maximum value of 100 interactions for training was defined.

The model output values are then denormalised, and to evaluate the network performance the RMS error (RMSE) was assumed, which considers the relationship between the predicted values obtained by the model, y_{predict} , and the observed data obtained through the data acquisition procedure, y_{obs} [37].

$$\text{RMSE} = \sqrt{\frac{1}{\eta} \sum_{i=1}^{\eta} (y_{\text{obs}} - y_{\text{predict}})^2}. \quad (12)$$

In addition, the coefficient of determination R^2 , which is based on the linear regression concerning the observed values, is also presented. In (13), $\overline{y_{\text{obs}}}$ represents the average of the observed data.

$$R^2 = 1 - \frac{\sum_{i=1}^{\eta} (y_{\text{obs}} - y_{\text{predict}})^2}{\sum_{i=1}^{\eta} (y_{\text{obs}} - \overline{y_{\text{obs}}})^2}. \quad (13)$$

Finally, the complete analysis considered 48,000 samples (η), which corresponds to the acquisition rate for a complete cycle of the ultrasonic equipment described in Section 3.2. Considering this signal record rate per second, it is possible to perform the complete ultrasonic data acquisition in just 1 s. Thus, the data matrices were built for both training and validation.

4.3 Feature extraction

For feature extraction, the wavelet energy coefficient (WEC) was assumed. The WEC takes information from both sides of the spectrum and can be obtained after the signal decomposition through wavelet packets transform (WPT), which is a generalisation of the wavelet transform (WT) [38]. WPT performs a new decomposition in each iteration and details the coefficients of previous iterations, the number of coefficients obtained depends on how many decompositions were performed [39].

In WPT, orthogonal wavelet decomposition is performed at various levels of WP nodes, forming a tree structure by dividing the approximation coefficients. The WP nodes define the level of WT filtering. After this division, an approximation coefficient vector and a detail coefficient vector are obtained. The information lost in the approximations is captured in these coefficients and a new vector is generated. In this technique, successive details are not reanalysed [40]. WP function can be described as follows:

$$W_{j,k}^n(t) = 2^{j/2} W^n(2^j t - k), \quad (14)$$

where j is the scale parameter, k is the translation parameter, and n represents the oscillation parameter. To perform the decomposition of the wavelet packages a Symlet function was used [41], which generates the scale filter coefficients for Daubechies less asymmetric wavelets. The Symlets are nearly symmetrical wavelets, those made discrete wavelet analysis practicable [42].

The first two WP functions $W_{0,0}^0(t) = \phi(t)$, $W_{0,0}^1(t) = \psi(t)$, for $n = 0$ and $n = 1$ represent the scale function and main function, respectively. Other functions for $n = 2, 3, \dots, N$ are defined according to:

$$\begin{aligned} W_{0,0}^{2n}(t) &= \sqrt{2} \sum_k h(k) W_{1,k}^n(2t - k), \\ W_{0,0}^{2n+1}(t) &= \sqrt{2} \sum_k g(k) W_{1,k}^n(2t - k). \end{aligned} \quad (15)$$

In (15), $h(k)$ is a low-pass filter and $g(k)$ is a high-pass filter. These functions are associated with a predefined scaling function and the mother wavelet function. The coefficients $\Omega_j^n(k)$ can be calculated using the function product $x(t)$ for each function of WP, according to the equation:

$$\Omega_j^n(k) = \int_{-\infty}^{\infty} x(t) W_{j,k}^n dt. \quad (16)$$

Each WP coefficient is specific, with a frequency resolution level associated with a scale parameter j , and an oscillation parameter n . The WT decomposes only the low-frequency components, while WPT decomposes all components at all levels and nodes. Finally, the WPT results in high- and low-frequency components. Concepts such as energy, entropy, and variation are used in the calculation of the coefficients of WP [21].

Energy is used to discriminate different classes since they contain the fault information and its fluctuation in a specific component that corresponds to the occurrence of a specific fault. The signal is decomposed into J levels, resulting in orthogonal subspaces, and accordingly, the frequency component can be calculated from the equation below:

$$E_J^n = \sum_k [\Omega_j^n(k)]^2. \quad (17)$$

For normalisation, in each frequency bandwidth, the distribution percentage of the power component is defined by:

$$e_J^n = \frac{E_J^n}{E_{\text{total}}} = \frac{E_J^n}{\sum_{n=1}^{2^J} E_J^n}. \quad (18)$$

The relative energy of the vector demonstrates the development of frequency over time in subspaces from low to high frequencies. To

obtain the best value from this package tree, the optimal binary value is determined. This is performed by calculating an optimal subtree, which is obtained according to the initial tree based on the entropy criterion. The resulting tree may be much smaller than the initial tree. In this way, the technique is applied considering that the objective is to find a minimum criterion for the algorithm, making it efficient [43].

The coefficients are allocated according to their Shannon entropy and are rebuilt to generate a filtered signal. Fig. 5 shows an example of the previously mentioned procedure considering 480 recorded points associated with the data acquisition experiment

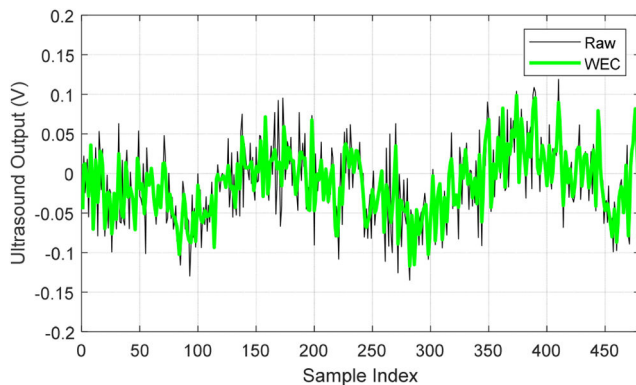


Fig. 5 Comparison between original and rebuild signal using WEC

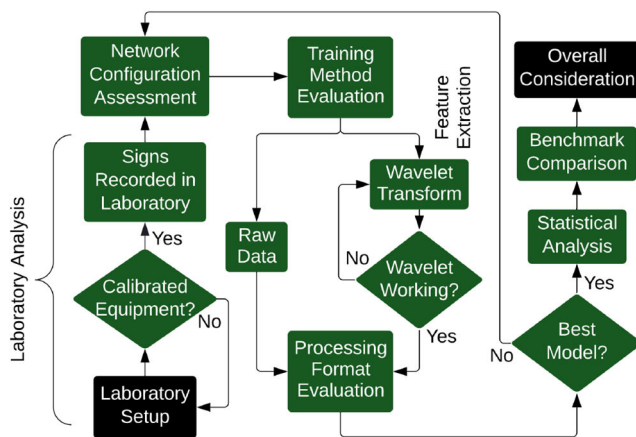


Fig. 6 Research development and evaluation flowchart

described in Section 3. This time series represents 10 ms, i.e. 1% of the evaluated signal, using two levels and one node.

In this case, coefficients can be assumed quantitatively to represent signal distributions combining their characteristics. Additionally, coefficients could be used efficiently for training when associated with a time series forecasting problem. For statistical evaluation of the model performance, the mean and standard deviation were computed.

The flowchart of the procedure performed in this article is shown in Fig. 6. Initially, the insulators were assembled in the laboratory and the ultrasound equipment was calibrated. After confirming that the ultrasonic signal was not saturated, the experiment signal was recorded. In the sequence, an analysis of the network configuration was performed to obtain the best parameters for the model. Then, the training method was assessed where the use of the raw signal was compared to the WPT method.

The objective was to determine the model parameters. Considering the best model found during the experiments mentioned above, both statistical analysis and comparisons considering two additional well-established models were reported.

5 Results and discussion

The analysis was initially performed based on the number of hidden units, network depth, and amount of data used for training. Results associated with these evaluations are presented in Table 1. This evaluation is performed using the SGDM training method, with raw data, using only the CPU, aiming to analyse the best network configuration.

In all tables presented in the sequence of this work, underlined results represent the best result in each category, and underlined and bold results are the overall best result. These initial configurations were used in this analysis as they require less computational effort.

The best RMSE results were obtained using 90% of the data for training with one layer. The use of more than one layer produced inferior results both in the RMSE calculation and convergence time.

Using 50 units, equivalent results were obtained in the RMSE calculation compared to 100 hidden units. However, the use of 50 hidden units was faster for convergence in most of the analysed cases and, this configuration was assumed for the next analysis. Fig. 7 presents the prediction results considering the previously mentioned configuration (50 hidden units) during the validation phase, which was compared to the original signal in a 480-point window. As can be verified, the model was able to capture the dynamic of the system and providing reduced noise results during the prediction of the ultrasound signal.

Table 1 Evaluation of network configuration

Number of Layers	Hidden Units	Train_test, %	RMSE, V	R^2	Time, s
1	50	50_50	0.0398	<u>0.6639</u>	582.435
		75_25	0.0399	0.6476	589.869
		90_10	<u>0.0392</u>	0.5653	<u>551.684</u>
1	100	50_50	0.0398	<u>0.6617</u>	<u>816.834</u>
		75_25	0.0399	0.6475	846.253
		90_10	<u>0.0392</u>	0.5706	932.011
2	50	50_50	0.0405	<u>0.6427</u>	829.117
		75_25	0.0405	0.6276	726.187
		90_10	<u>0.0397</u>	0.5408	<u>646.201</u>
2	100	50_50	0.0403	<u>0.6523</u>	<u>1361.890</u>
		75_25	0.0405	0.6293	1685.749
		90_10	<u>0.0397</u>	0.5481	1939.361
3	50	50_50	0.0404	<u>0.6366</u>	<u>1108.712</u>
		75_25	0.0406	0.6303	1182.944
		90_10	<u>0.0399</u>	<u>0.5426</u>	1269.466
3	100	50_50	0.0410	0.6027	<u>2136.133</u>
		75_25	0.0410	<u>0.6142</u>	2602.330
		90_10	<u>0.0401</u>	0.5218	2945.282

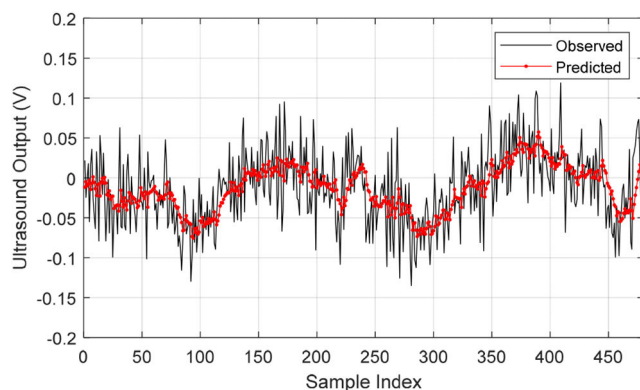


Fig. 7 Forecast against the observed signals

Table 2 Evaluation of network with WEC

Levels	Nodes	RMSE, V	R^2	Time, s
2	1	0.0225	0.8465	358.457
	2	0.0115	0.9014	383.963
	3	<u>0.0098</u>	<u>0.9691</u>	393.171
3	1	0.0223	0.8486	<u>383.158</u>
	2	0.0121	0.8866	401.058
	3	0.0077	<u>0.9802</u>	451.586
4	1	0.0237	0.8308	419.046
	2	0.0117	0.8978	477.773
	3	0.0077	0.9805	<u>388.474</u>
5	1	0.0223	0.8523	382.583
	2	0.0107	0.9157	<u>378.151</u>
	3	<u>0.0083</u>	<u>0.9778</u>	385.313

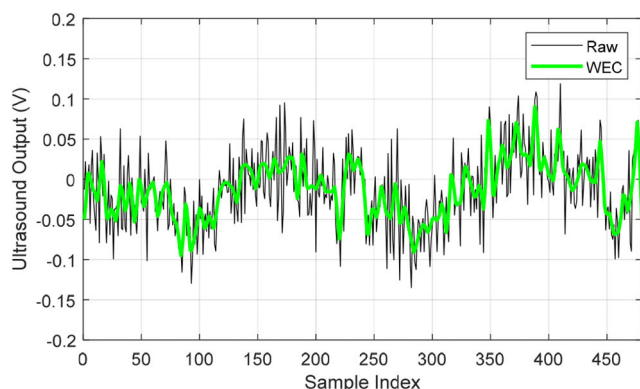


Fig. 8 Best WEC model for LSTM analysis compared to the original signal: four levels and three nodes

The application of the WEC for feature extraction and noise reduction compared to the raw signal with different configurations is presented in Table 2.

Even using more levels of the package tree, it was not possible to significantly improve the predictive capability of the LSTM wavelet. However, assuming more nodes, significant improvement in the network predictability with LSTM wavelet was obtained.

Assuming >3 nodes, the signal loses its characteristics in some analysed windows, thus the best RMSE result, within an acceptable training time, maintaining signal characteristics was obtained using four levels and three nodes at WEC. Fig. 8 shows the result of using this WEC configuration compared to the original signal.

Table 3 shows the application of WEC compared to the original signal when the LSTM wavelet method was applied. Also, different training methods and variations in the form of data processing are presented.

The use of the WEC yielded promising results concerning the RMSE calculation. The use of the WEC to reduce chaotic time series noise proved to be an interesting technique to be associated with LSTM.

The ADAM training method provided better RMSE and R^2 results than other methods. There was little difference between CPU, GPU, and AUTO mode for RMSE.

In this case, ADAM with AUTO mode was assumed as a better result because it provided reasonable speed training with smaller error and high accuracy. Fig. 9 can be seen as the application of this model. Table 3 shows that there is no considerable difference in the use of the GPU to analyse the problem in question. Thus, a large investment in GPU is not justified to develop the study using the proposed model.

According to the results presented in this paper, the WT improves the signal forecast by reducing signal noise. There is a great difference in accuracy when the algorithm was evaluated without the use of adequate filters to reduce the noise, as it can be seen in Table 3.

Comparing with classic algorithms as performed for the benchmarking presented in Tables 4 and 5, the proposed algorithm has better accuracy in all variations of the model that were evaluated in this study. These results are also validated in comparison with other studies, as presented in [14, 15, 16, 17], studies reporting that the LSTM was superior to the SVM algorithm in similar time series forecasting applications as the one presented in this research.

5.1 Statistical analysis

For a complete statistical evaluation of the time series forecasting approach, we considered ten runs with the best model found for this purpose. This analysis was performed assuming the four data sets of 48,000 points, which corresponds to different time windows within the whole time series.

The complete assessment of the algorithm's prediction is present in Table 6. In this analysis, the mean RMSE mean R^2 , mean time, standard deviation, and variance are presented. Even changing the data set, RMSE values remained small. These values demonstrate the robustness of the algorithm in dealing with different data sets.

5.2 Benchmark analysis

Assuming the task of comparing the wavelet LSTM with well-established methods for time series forecasting, in this section, it was considered two more structures: a non-linear AutoRegressive (NAR) model, and a NAR with exogenous input (NARX) model.

Considering the scaled conjugate gradient, Bayesian regularisation and Levenberg–Marquardt (LM), the variation of the training method did not significantly impact the final results of the models. In this way, we present only the results using the LM training method. Additionally, distinct configuration parameters were assumed, they are the number of hidden neurons (HNs) and the number of delayed outputs. In the NAR network, the calculation is based on data set 1, and in NARX networks the calculation is based on the data relationship of data set 1 to data set 2. Data set 2 represents values in a time window ahead of data set 1.

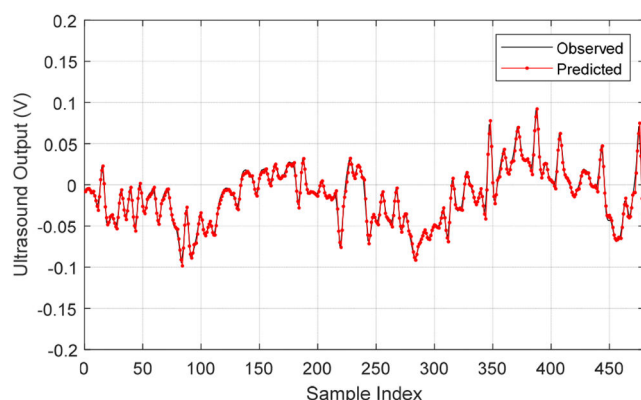
Tables 4 and 5 show the benchmark for the two methods described above. Results are presented for network evaluation, considering 90% of the data was used for training, 5% for validation, and 5% for testing. In this benchmark, the analysis configuration was performed considering the number of HN and the number of regressors (NR).

The NAR and NARX model provided lower prediction accuracy when compared to the proposed Wavelet LSTM approach. Without combining LSTM with Wavelet, the obtained results were similar to both NAR and NARX models.

In the best case of the NAR model, it reached an RMSE equal to 0.0319, which is a bigger value than wavelet LSTM, which reported 0.0047 in the best case. The variability obtained when the results of both NAR and NARX models were analysed are higher than the wavelet LSTM method.

Table 3 Evaluation of training method and model

Training Method	Data format	Data proc.	RMSE, V	R^2	Time, s
SGDM	raw	CPU	<u>0.0391</u>	0.5650	<u>388.507</u>
		GPU	0.0392	0.5579	502.951
		AUT	<u>0.0391</u>	<u>0.5675</u>	532.422
RMSP	raw	CPU	0.0363	0.6837	<u>387.612</u>
		GPU	0.0347	0.6821	537.124
		AUT	<u>0.0342</u>	<u>0.7005</u>	528.446
ADAM	raw	CPU	<u>0.0332</u>	0.7309	388.710
		GPU	0.0333	0.7204	533.006
		AUT	<u>0.0333</u>	<u>0.7311</u>	541.788
SGDM	WEC	CPU	<u>0.0072</u>	<u>0.9835</u>	386.494
		GPU	0.0079	0.9801	543.585
		AUT	0.0083	0.9772	507.112
RMSP	WEC	CPU	0.0058	0.9861	<u>398.674</u>
		GPU	<u>0.0055</u>	<u>0.9910</u>	519.831
		AUT	0.0060	0.9858	517.401
ADAM	WEC	CPU	0.0049	0.9924	<u>526.733</u>
		GPU	0.0048	0.9930	535.671
		AUT	0.0047	0.9931	535.730

**Fig. 9** Best model for time series forecasting: ADAM training and CPU mode**Table 4** Benchmark evaluation with NAR model

HN_NR	RMSE, V	R^2	Std deviation	Variance
05_05	0.03676	0.7597		
05_10	0.03750	0.7919		
05_15	0.03289	0.8172		
10_05	0.03784	0.7554		
10_10	0.04377	0.7926	4.27×10^{-3}	1.82×10^{-5}
10_15	0.03186	0.8236		
15_05	0.03832	0.7451		
15_10	0.03498	0.7803		
15_15	0.03251	0.8187		

Table 5 Benchmark evaluation with the NARX model

HN_NR	RMSE, V	R^2	Std deviation	Variance
05_05	0.03791	0.7411		
05_10	0.03778	0.7918		
05_15	0.03248	0.8199		
10_05	0.03726	0.7451		
10_10	0.04461	0.7806	3.73×10^{-3}	1.39×10^{-5}
10_15	0.03210	0.8275		
15_05	0.03740	0.7391		
15_10	0.04182	0.7943		
15_15	0.03265	0.8190		

6 Conclusion

The contributions of this paper for electric distribution power systems research are summarised in the following: The first contribution is focused on improving electrical power grid inspection. Automating field inspection using AI models can assist operators in identifying failures, thereby increasing the reliability of the electrical power grid; the second contribution is related to finding an algorithm with greater accuracy using the WT. Through the use of WEC, it is possible to reduce noise and extract the feature of the signal thus improving prediction analysis; the third contribution is about improving the quality of predictive maintenance to reduce the need for corrective maintenance in the system. Through a reliable forecasting technique, it is possible to identify a fault before it occurs and thereby improve team management.

The use of LSTM showed good predictability, even considering a noisy chaotic time series, for the insulator condition time series prediction problem. By assuming WEC, it was possible to considerably improve the signal predictability. More nodes in the WPT tree calculation reduced the signal noise for WEC, however, when an increased number of nodes was assumed, depending on the signal characteristic, there may be a loss of relevant information. In this way, making the prediction was unreliable. The application of the WT could improve the predictability associated with this problem, but a specific model should be assessed for each case. The use of the WT proved to be a very promising technique to be used with the prediction of time series in conjunction with LSTM, distinct mother wavelets can be applied for future research of this problem.

In this evaluation, the use of AUTO mode improved some analysis. Using more data for training generates a smaller error by improving the prediction assessment. The use of autoregressive non-linear algorithms yields higher RMSE and lower accuracy compared to the proposed wavelet LSTM method, nevertheless with little variance. The proposed method can be used differentiating insulator conditions stages of insulator degradation, from the signal forecast and knowing where there will be contamination that may develop a failure, it will be possible to predict the failure before it occurs.

One of the advantages of using ultrasound equipment is its ability to be a non-destructive technique that can be applied without the proximity of the electrical power system, thus being safer for field inspection. Using ultrasound, it is possible to evaluate the insulators' condition instead of traditional approaches for prediction of leakage current events. The proposed strategy showed to be a promising method for inspection of electric power utilities.

Table 6 Statistical evaluation

Mean RMSE, V	Mean R^2	Mean time, s	Std. dev.	Variance
0.0047	0.9930	785.334	1.08×10^{-4}	1.17×10^{-8}
0.0049	0.9932	808.881	9.43×10^{-5}	8.89×10^{-9}
0.0052	0.9967	842.439	1.29×10^{-4}	1.66×10^{-8}
0.0053	0.9968	516.042	1.07×10^{-3}	1.15×10^{-6}

As discussed in the previous subsection, Wavelet LSTM provided a higher accuracy compared to other algorithms such as non-linear regressive networks, this statement is also supported by the related results of [13–17]. In most studies focusing on fault identification of insulators, the algorithm evaluates only performance in terms of accuracy. In this article, it has been shown that LSTM can be improved by applying a proper configuration to the problem and extracting features through WEC. This work showed that the software can be successfully applied to forecast chaotic time series, in this way it will be possible to develop a hardware project in the future for analysis with the embedded algorithm. Currently, techniques using deep learning have high computational effort, which makes the development of embedded algorithms for online applications a challenge.

7 References

- [1] Anteneh, D., Khan, B.: 'Reliability enhancement of distribution substation by using network reconfiguration a case study at debre berhan distribution substation', *Int. J. Econ. Energy Environ.*, 2019, **4**, (2), pp. 33–40
- [2] Khan, B., Alhelou, H.H., Mebrahtu, F.: 'A holistic analysis of distribution system reliability assessment methods with conventional and renewable energy sources', *AIMS Energy*, 2019, **7**, (4), pp. 413–429
- [3] Dalgitsis, A., Louta, M., Karetos, G.T.: 'Traffic forecasting in cellular networks using the LSTM RNN'. 2018 Proc. of Pan-Hellenic Conf. on Informatics (PCI'18), New York, USA, November 2018, pp. 28–33
- [4] Jiang, Z., Jiang, X., Guo, Y., et al.: 'Pollution accumulation characteristics of insulators under natural rainfall', *IET Gener. Transm. Distrib.*, 2017, **11**, (6), pp. 1479–1485
- [5] Stefenon, S.F., Américo, J.P., Meyer, L.H., et al.: 'Analysis of the electric field in porcelain pin-type insulators via finite elements software', *IEEE Latin Am. Trans.*, 2019, **16**, (10), pp. 2505–2512
- [6] Stefenon, S.F., Oliveira, J.R., Coelho, A.S., et al.: 'Diagnostic of insulators of conventional grid through LabVIEW analysis of FFT signal generated from ultrasound detector', *IEEE Latin Am. Trans.*, 2017, **15**, (5), pp. 884–889
- [7] Ibrahim, A., Dalbah, A., Abualsaud, A., et al.: 'Application of machine learning to evaluate insulator surface erosion', *IEEE Trans. Instrum. Meas.*, 2020, **69**, (2), pp. 314–316
- [8] Abouzeid, A.K., El-Hag, A., Assaleh, K.: 'Equivalent salt deposit density prediction of silicone rubber insulators under simulated pollution conditions', *Electr. Power Compon. Syst.*, 2018, **46**, (10), pp. 1123–1133
- [9] Polisetty, S., El-Hag, A., Jayram, S.: 'Classification of common discharges in outdoor insulation using acoustic signals and artificial neural network', *High Volt.*, 2019, **4**, (4), pp. 333–338
- [10] Stefenon, S.F., Silva, M.C., Bertol, D.W., et al.: 'Fault diagnosis of insulators from ultrasound detection using neural networks', *J. Intell. Fuzzy Syst.*, 2019, **37**, (5), pp. 6655–6664
- [11] Corso, M.P., Stefenon, S.F., Couto, V.F., et al.: 'Evaluation of methods for electric field calculation in transmission lines', *IEEE Latin Am. Trans.*, 2018, **16**, (12), pp. 2970–2976
- [12] Li, X., Yu, C., Su, F., et al.: 'Novel training algorithms for long short-term memory neural network', *IET Signal Process.*, 2019, **13**, (3), pp. 304–308
- [13] Zhao, H., Sun, S., Jin, B.: 'Sequential fault diagnosis based on LSTM neural network', *IEEE Access*, 2018, **6**, pp. 12929–12939
- [14] Lu, W., Cheng, Y., Meng, D., et al.: 'Early fault detection approach with deep architectures', *IEEE Trans. Instrum. Meas.*, 2018, **67**, (7), pp. 1679–1689
- [15] Bhattacharya, B., Sinha, A.: 'Intelligent fault analysis in electrical power grids'. 2017 IEEE 29th Int. Conf. on Tools with Artificial Intelligence (ICTAI), Boston, USA, November 2017, pp. 985–990
- [16] Zhang, S., Wang, Y., Liu, M., et al.: 'Data-based line trip fault prediction in power systems using LSTM networks and SVM', *IEEE Access*, 2018, **6**, pp. 7675–7686
- [17] Xia, Y., Lu, J., Li, H., et al.: 'A deep learning based image recognition and processing model for electric equipment inspection'. 2018 2nd IEEE Conf. on Energy Internet and Energy System Integration (EI2), Beijing, China, October 2018, pp. 1–6
- [18] Cheng, L., Mei, H., Wang, L., et al.: 'Research on aging evaluation and remaining lifespan prediction of composite insulators in high temperature and humidity regions', *IEEE Trans. Dielectr. Electr. Insul.*, 2016, **23**, (5), pp. 2850–2857
- [19] Ali, H.: 'Leakage current prediction for high voltage insulators flashover based on extreme value theory'. 2016 Int. Symp. on Computer, Consumer and Control (IS3C), Xi'an China, August 2016, pp. 870–873
- [20] Wang, H., Jin, L., Lu, P., et al.: 'Physical stages dividing and quantitative study of insulator surface discharges'. 2018 28th Int. Symp. on Discharges and Electrical Insulation in Vacuum (ISDEIV), Greifswald Germany, November 2018, pp. 131–134
- [21] Alves, D.K., Costa, F.B., Ribeiro, R.L.A., et al.: 'Real-time power measurement using the maximal overlap discrete wavelet-packet transform', *IEEE Trans. Ind. Electron.*, 2017, **64**, (4), pp. 3177–3187
- [22] Stefenon, S.F., Steinheuser, D.F., Silva, M.P., et al.: 'Application of active methodologies in engineering education through the integrative evaluation at the universidade do planalto catarinense, Brazil', *Interiencia*, 2019, **44**, (7), pp. 408–413
- [23] Stefenon, S.F., Branco, N.W., Nied, A., et al.: 'Analysis of training techniques of ANN for classification of insulators in electrical power systems', *IET Gener. Transm. Distrib.*, 2020, **14**, (8), pp. 1591–1597
- [24] He, H., Luo, D., Lee, W., et al.: 'A contactless insulator contamination levels detecting method based on infrared images features and RBFNN', *IEEE Trans. Ind. Appl.*, 2019, **55**, (3), pp. 2455–2463
- [25] Kin, T., Jeon, S., Lee, Y., et al.: 'Three-dimensional computed tomography and composition analysis of porcelain insulators for 154 kV power transmission lines', *IEEE Trans. Dielectr. Electr. Insul.*, 2019, **26**, (1), pp. 115–119
- [26] Kalla, U.K., Suthar, R., Sharma, K., et al.: 'Power quality investigation in ceramic insulator', *IEEE Trans. Ind. Appl.*, 2018, **54**, (1), pp. 121–134
- [27] Fresno, J.M., Ardilla-Rey, J.A., Martinez-Tarifa, J.M., et al.: 'Partial discharges and noise separation using spectral power ratios and genetic algorithms', *IEEE Trans. Dielectr. Electr. Insul.*, 2017, **24**, (1), pp. 31–38
- [28] Anjum, S., Jayaram, S., El-Hag, A., et al.: 'Detection and classification of defects in ceramic insulators using RF antenna', *IEEE Trans. Dielectr. Electr. Insul.*, 2017, **24**, (1), pp. 183–190
- [29] Park, K., Motai, Y., Yoon, J.R.: 'Acoustic fault detection technique for high-power insulators', *IEEE Trans. Ind. Electron.*, 2017, **64**, (12), pp. 9699–9708
- [30] Stefenon, S.F., Ribeiro, M.H.D.M., Nied, A., et al.: 'Wavelet group method of data handling for fault prediction in electrical power insulators', *Int. J. Electr. Power Energy Syst.*, 2020, **123**, p. 106269
- [31] Stefenon, S.F., Grebogi, R.B., Freire, R.Z., et al.: 'Optimized ensemble extreme learning machine for classification of electrical insulators conditions', *IEEE Trans. Ind. Electron.*, 2020, **67**, (6), pp. 5170–5178
- [32] Pan, J., Zi, Y., Chen, J., et al.: 'Liftingnet: A novel deep learning network with layerwise feature learning from noisy mechanical data for fault classification', *IEEE Trans. Ind. Electron.*, 2018, **65**, (6), pp. 4973–4982
- [33] Motepi, S., Hasan, A.N., Stopforth, R.: 'Improving load forecasting process for a power distribution network using hybrid AI and deep learning algorithms', *IEEE Access*, 2019, **7**, pp. 82584–82598
- [34] MATLAB (Version R2020): 'Long short-term memory networks', Available at <https://www.mathworks.com/help/deeplearning/ug/long-short-term-memory-networks.html>
- [35] Sagheer, A., Kotb, M.: 'Time series forecasting of petroleum production using deep LSTM recurrent networks', *Neurocomputing*, 2019, **323**, pp. 203–213
- [36] Vani, S., Rao, T.V.M.: 'An experimental approach towards the performance assessment of Various optimizers on convolutional neural network'. 2019 3rd Int. Conf. on Trends in Electronics and Informatics (ICOEI), Tirunelveli India, October 2019, pp. 331–336
- [37] Kasburg, C., Stefenon, S.F.: 'Deep learning for photovoltaic generation forecast in active solar trackers', *IEEE Lat. Am. Trans.*, 2019, **17**, (12), pp. 2013–2019
- [38] Hafiz, F., Swain, A., Naik, C., et al.: 'Identification of power quality events: selection of optimum base wavelet and machine learning algorithm', *IET Sci. Meas. Technol.*, 2019, **13**, (2), pp. 260–271
- [39] Stefenon, S.F., Freire, R.Z., Coelho, L.S., et al.: 'Electrical insulator fault forecasting based on a wavelet neuro-fuzzy system', *Energies*, 2020, **13**, p. 484
- [40] He, W., He, Y., Li, B., et al.: 'Feature extraction of analogue circuit fault signals via cross-wavelet transform and variational Bayesian matrix factorization', *IET Sci. Meas. Technol.*, 2019, **13**, (2), pp. 318–327
- [41] Vonesch, C., Blu, T., Unser, M.: 'Generalized daubechies wavelet families', *IEEE Trans. Signal Process.*, 2007, **55**, (9), pp. 4415–4429
- [42] Siddiqi, M.H., Sulaiman, S.B.T., Faye, I., et al.: 'A real time specific weed discrimination system using multi-level wavelet decomposition', *Int. J. Agric. Biol.*, 2009, **11**, (5), pp. 559–565
- [43] Ribeiro, M.H.D.M., Stefenon, S.F., de Lima, J.D., et al.: 'Electricity price forecasting based on self-adaptive decomposition and heterogeneous ensemble learning', *Energies*, 2020, **13**, (19), p. 5190